BUSINESS INTELLIGENCE REPORT

Food Delivery Platform

Analytics & Operational Insights

01 / EXECUTIVE SUMMARY

Executive Summary

This report presents a comprehensive analysis of a food delivery platform's operational and commercial data. The primary objective was to translate raw transactional data into actionable business intelligence that can directly inform decisions across revenue growth, operational efficiency, and customer experience.

The analysis was conducted using Microsoft Power BI, encompassing data modelling, DAX measure development, and multi-page interactive dashboard design. The dataset covers 48,200+ orders, spanning delivery zones, food categories, customer segments, restaurant partners, and time-series trends.

Key Findings at a Glance

48,200+ Total Orders Analysed	**₹6.4M** Total Platform Revenue	**34.2 min** Avg Delivery Time
3.84 / 5 Average Customer Rating	**50%** Repeat Customer Rate	**48%** Late Delivery Rate

Problem Statement

Food delivery platforms operate at the intersection of three complex systems: customer demand (inherently variable), restaurant supply (capacity-constrained), and last-mile logistics (geographically fragmented). Understanding how these systems interact — and where friction costs the business money — requires more than surface-level reporting.

The business questions driving this analysis were:

- Where are operational inefficiencies impacting delivery performance?
- What factors influence customer retention and repeat behaviour?
- How does performance vary across cities and restaurants?
- Which areas require optimization for better service quality?
- How is revenue distributed across restaurants and categories?

Data & Methodology

Data Model Architecture

The Power BI data model was built on a star schema — a best-practice structure for analytical workloads that optimises both query performance and DAX formula complexity.

Table	Type	Records	Key Fields
Orders (Fact)	Fact	48,200+	OrderID, RestaurantID, CustomerID, ZoneID, DateTime
Customers (Dim)	Dimension	~22,000	CustomerID, Segment, JoinDate, Location
Restaurants (Dim)	Dimension	~480	RestaurantID, Category, ZoneID, AvgRating
Zones (Dim)	Dimension	4	ZoneID, ZoneName, AvgDistanceKm
Date (Dim)	Date Table	365	Date, Week, Month, Quarter, IsWeekend
Categories (Dim)	Dimension	12	CategoryID, CategoryName, PriceSegment

DAX Measures (Selected)

Measure	DAX Logic	Use
Total Revenue	SUMX(Orders, Orders[Quantity] * Orders[UnitPrice])	Core revenue KPI
Avg Delivery Time	AVERAGEX(Orders, Orders[DeliveryMins])	Operations KPI
SLA Breach Rate	DIVIDE(COUNTROWS(FILTER(Orders, [DeliveryMins]>35)), COUNTROWS(Orders))	SLA monitoring
Repeat Customer %	DIVIDE([RepeatCustomers],[TotalUniqueCustomers])	Retention KPI
Conversion Rate	DIVIDE([OrdersPlaced],[AppVisits])	Funnel KPI
Revenue MoM %	DIVIDE([RevenueCurrent]-[RevenuePriorMonth],[RevenuePriorMonth])	Trend analysis

Key Business Insights

Insight 1: High Delivery Delay Rate

A significant portion of orders are delivered late (~48%), indicating operational inefficiencies in delivery processes.

Impact: HIGH — affects customer satisfaction and retention

Insight 2 Geographic Performance Variation

Delivery performance varies across cities, with some locations showing higher delays and cancellations.

Impact: HIGH — Operational efficiency & cancellations

Insight 3: Strong Repeat Customer Contribution

Approximately half of the customers are repeat users, contributing significantly to total order volume.

Impact: HIGH — retention and revenue stability

Insight 4 Revenue Concentration in Top Restaurants

A small group of restaurants contributes a large share of total revenue, indicating dependency on key partners.

Impact: Medium — business risk

Insight 5: Stable Order & Revenue Trends

Order and revenue trends show consistent patterns over time, indicating steady platform activity.

Impact: Medium — business stability

Strategic Recommendations

Each recommendation below is directly traceable to a data-backed insight. Estimated impact ranges are directional based on analogous platform benchmarks and intra-dataset modelling — they should be validated against live A/B tests.

R1 — Dynamic Demand Management & Pre-Scheduling

Implement demand-based delivery fee adjustments during peak windows (surge pricing) combined with pre-order discounts. Users who pre-schedule orders for 30–60 minutes ahead receive a ₹20–30 discount; the platform benefits from flattened demand curves, improved rider utilisation, and protected delivery SLAs.

- reduction in peak-hour SLA breaches
- increase in off-peak revenue

R2 — Strengthen Customer Retention

Introduce targeted loyalty programs and personalized offers to encourage repeat purchases. Focus on engaging existing customers through incentives based on their order history and preferences.

- Increase in repeat customer rate
- Improved customer retention
- Higher order frequency per user

R3 — Optimize City-Level Operations

Identify cities with high delivery delays and operational inefficiencies, and implement localized strategies such as better resource allocation and optimized delivery planning.

- Reduction in city-level delivery delays
- Improved operational performance across regions
- Better service consistency

R4 — Enhance Customer Experience

Improve overall customer experience by increasing transparency in delivery status and minimizing delays. Providing clear communication throughout the order journey can improve satisfaction.

- Improved customer satisfaction
- Reduction in complaints and cancellations
- Enhanced user trust

R5 — Reduce Dependency on Top Restaurants

Promote mid-performing restaurants and expand the diversity of restaurant partners to balance revenue distribution and reduce reliance on a few high-performing vendors.

- More balanced revenue contribution
- Reduced business risk
- Increased platform variety

Tools & Technologies

Tool / Technology	Role in Project
Microsoft Power BI Desktop	Primary analytics and visualisation environment
Power Query (M Language)	ETL — data extraction, cleaning, transformation
DAX (Data Analysis Expressions)	All KPI measures, calculated columns, time intelligence
Star Schema Data Modelling	Relational model design for analytical workloads
Custom Visuals	Heatmap, advanced gauge, funnel chart
Report Theming (JSON)	Consistent colour palette and typography across all pages
Bookmarks & Drill-through	Interactive UX for stakeholder self-service exploration
Row-Level Security (RLS)	Zone-based data access control

Skills Demonstrated

- Business problem decomposition — translating ambiguous questions into analytical frameworks
- Data modelling — star schema design, relationship management, and cardinality control
- Advanced DAX — time intelligence, context transition, CALCULATE, FILTER, iterators
- Data storytelling — structuring insights for non-technical stakeholder audiences
- Visual design — accessibility, hierarchy, and colour semantics in dashboard layouts
- Operational analytics — KPI design, SLA monitoring, and threshold-based alerting

Conclusion

This project demonstrates the end-to-end application of business intelligence skills to a real-world operational context. The food delivery domain is analytically rich — combining high-frequency transactional data, geographic complexity, multi-sided marketplace dynamics, and customer behaviour patterns — making it an ideal environment to showcase the full analytical stack.